

Mini RFC: Connection Multiplexing

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Note: This is a simplified and annotated summary of the QUIC-related IETF RFCs. It's part of a course project at Tsinghua University and not suitable for any other purposes.

We reorder paragraphs in QUIC RFCs and omit many details to improve its readability for beginners.

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1 Overview

Incoming packets are classified on receipt. Packets can either be associated with an existing connection or – for servers – potentially create a new connection.

Endpoints try to associate a packet with an existing connection. If the packet has a non-zero-length Destination Connection ID corresponding to an existing connection, QUIC processes that packet accordingly. Note that more than one connection ID can be associated with a connection.

If the Destination Connection ID is zero length and the addressing information in the packet matches the addressing information the endpoint uses to identify a connection with a zero-length connection ID, QUIC processes the packet as part of that connection. An endpoint can use just destination IP and port or both source and destination addresses for identification, though this makes connections fragile.

Endpoints can send a Stateless Reset for any packets that cannot be attributed to an existing connection. A Stateless Reset allows a peer to more quickly identify when a connection becomes unusable.

Packets that are matched to an existing connection are discarded if the packets are inconsistent with the state of that connection. For example, packets are discarded if they indicate a different protocol version than that of the connection or if the removal of packet protection is unsuccessful once the expected keys are available.

Invalid packets that lack strong integrity protection, such as Initial, Retry, or Version Negotiation, MAY be discarded. An endpoint MUST generate a

connection error if processing the contents of these packets prior to discovering an error, or fully revert any changes made during that processing.

2 Client Packet Handling

Valid packets sent to clients always include a Destination Connection ID that matches a value the client selects. Clients that choose to receive zero-length connection IDs can use the local address and port to identify a connection. Packets that do not match an existing connection – based on Destination Connection ID or, if this value is zero length, local IP address and port – are discarded.

Due to packet reordering or loss, a client might receive packets for a connection that are encrypted with a key it has not yet computed. The client MAY drop these packets, or it MAY buffer them in anticipation of later packets that allow it to compute the key.

If a client receives a packet that uses a different version than it initially selected, it MUST discard that packet.

3 Server Packet Handling

If a server receives a packet that indicates an unsupported version and if the packet is large enough to initiate a new connection for any supported version, the server SHOULD send a Version Negotiation packet. A server MAY limit the number of packets to which it responds with a Version Negotiation packet. Servers MUST drop smaller packets that specify unsupported versions.

The first packet for an unsupported version can use different semantics and encodings for any version-specific field. In particular, different packet protection keys might be used for different versions. Servers that do not support a particular version are unlikely to be able to decrypt the payload of the packet or properly interpret the result. Servers SHOULD respond with a Version Negotiation packet, provided that the datagram is sufficiently long.

Packets with a supported version, or no Version field, are matched to a connection using the connection ID or – for packets with zero-length connection IDs – the local address and port. These packets are processed using the selected connection; otherwise, the server continues as described below.

If the packet is an Initial packet fully conforming with the specification, the server proceeds with the handshake. This commits the server to the version that the client selected.

If a server refuses to accept a new connection, it SHOULD send an Initial packet containing a CONNECTION_CLOSE frame with error code CONNECTION_REFUSED.

If the packet is a 0-RTT packet, the server MAY buffer a limited number of these packets in anticipation of a late-arriving Initial packet. Clients are

not able to send Handshake packets prior to receiving a server response, so servers SHOULD ignore any such packets.

Servers MUST drop incoming packets under all other circumstances.